

Sathyaseelan K

Karur, Tamil Nadu, India • +91 97904 19564 • ksathyaseelan34@gmail.com
github.com/sathyaseelan2006 • linkedin.com/in/sathyaseelan-dev • sathya34.vercel.app

Summary

Final-year IT undergraduate with hands-on experience building scalable backend systems, REST APIs, and full-stack applications across Java, Python, and the Supabase stack. Seeking Software Engineering / Backend roles building scalable, data-driven solutions.

Education

Bachelor of Technology in Information Technology
M. Kumarasamy College of Engineering, Karur, TN
Aug 2023 – Present • CGPA: 7.9/10.0

Higher Secondary Certificate (HSC)
Chelammal Vidhyaashram, Trichy, TN
Jun 2022 – Jun 2023 • Score: 86%

Technical Skills

Programming Languages	Java (Proficient), Python (Intermediate), SQL (Proficient), C (Basic)
Web Technologies	HTML, CSS, Flask
Databases & Backend	MySQL, Supabase, MongoDB
Core Subjects	Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
AI / ML	Machine Learning, Computer Vision, OpenCV, TensorFlow, NLP
Tools & Platforms	Git, GitHub, Postman, IntelliJ IDEA, VS Code, Android Studio, Google Cloud Platform

Experience

Google Gemini Student Ambassador May 2026 – Present
Google Campus Ambassador Program

- Spearheaded campus-wide adoption of Google's Gemini AI by organizing technical workshops and coding sessions, empowering students to build AI-integrated applications.
- Mentored peers on prompt engineering, API integrations, and leveraging Large Language Models (LLMs) to optimize modern software development workflows.

Projects

PolyVault – Lightweight Knowledge Management & File Vault Platform ([Live](#) | [Code](#)) May – June 2026
Java, HttpServer, ServerSocket, Firebase Auth, JWT, SVG, GZIP

- Built a dependency-free knowledge-management and file-vault platform with a custom backend on Java's native `HttpServer` and `ServerSocket` APIs.
- Implemented Firebase Auth, JWT validation, and GZIP-compressed versioned storage, plus an interactive SVG mind-map visualization across Web and CLI clients.

IoT-Based Active Peripheral Security and Automated Attendance ([Code](#)) Feb – Mar 2026
ESP32, RC522 RFID, INA219, Flutter, MySQL

- Engineered an ESP32-based hardware framework using RC522 RFID and a 5V relay circuit to instantly block unauthorized peripheral access, successfully securing physical endpoints.
- Built a cross-platform Flutter app and MySQL database for live monitoring, completely replacing manual attendance registers with 100% automated, tamper-proof digital records.

Megam – AI-Powered Global Air Quality Monitoring Platform ([Live](#) | [Code](#)) Nov – Dec 2025
React, TypeScript, Flask, Python, scikit-learn, NASA MODIS

- Engineered a Python/Flask data pipeline integrating IQAir and NASA MODIS APIs to successfully aggregate and process live metrics across 80,000+ monitoring stations.
- Developed scikit-learn predictive models and a React/TypeScript front-end with a Three.js 3D globe, enabling users to interactively visualize and forecast pollution trends globally.

Re:Vive – AI, NLP and AR Cultural Discovery Platform ([Live](#) | [Code](#)) Jun – Dec 2025
Flask, NLP, Supabase, JavaScript, AR Integration

- Architected a Flask-based backend utilizing NLP techniques to generate personalized landmark recommendations and multilingual narration, significantly enhancing cultural accessibility.
- Integrated AR-based discovery features backed by Supabase; the system's architecture and findings were peer-reviewed and published at the IEEE ICCCES 2026 conference.

Research Publication

- IEEE Publication: "[Re:Vive: An Intelligent Interactive Platform for Cultural Discovery through AI, NLP, and AR](#)" – ICCCES 2026.

Certifications

- Oracle Certified: Data Science Professional; Cloud Database Services (Oct 2025).
- Coursera: Supervised Machine Learning – DeepLearning.AI & Stanford Online (Sep 2025).
- Coursera: Cybersecurity for Everyone – University of Maryland (Feb 2025).